

LAB MODULE #1: REACTOR SYSTEMS WALKDOWN – PART ACONTROL ROOM – REACTOR CONSOLE

- NFMS STARTUP CHANNEL – Fission chamber; < 2CPS trip, >9E4 Trip, recorder trace
- LOG & LINEAR ('LOG-N') CHANNEL – CIC; < 4 watt trip; 150kW flow/flapper SCRAM enable; recorder trace
- LINEAR POWER CHANNEL – CIC; 1.10MW reverse; 1.2 MW Scram; recorder trace
- SAFETY CHANNEL - UIC; 150kW flow/flapper SCRAM enable; 1.2 MW Scram; recorder trace
- N-16 CHANNEL – Xenon filled Ion Chamber; power calibration
- CONTROL ROD INDICATORS – drive switches; ON gang; up/down/ off mag lights
- SCRAM LOGIC UNIT - Seven (7) SCRAM's (*reference diagram*):
 - o Linear Power Channel $\geq 120\%$ Full Power (1MW)
 - o Safety Power Channel $\geq 120\%$ Full Power (1MW)
 - o Primary Coolant Flow ≤ 475 GPM or ≥ 525 GPM (if power ≥ 150 kW)
 - o Flapper Valve OPEN (if power ≥ 150 kW)
 - o Pool Level ≤ 36 " below full
 - o Pool Temperature $T_2 \geq 116^\circ\text{F}$
 - o Manual – Red Button
 - Plus:
 - Console Keyswitch
 - SCRAM Logic Ground Fault
- TEMPERATURE MONITORING (*reference primary system schematic*):
 - o T-2: Pool
 - o T-3: Cold Primary Pool Return
 - o T-4: Hot Primary Pool Outlet (from core)
 - o T-5: Heat Exchanger – Hot Primary Inlet
 - o T-6: Heat Exchanger – Cold Primary Outlet
 - o T-7: Heat Exchanger – Cold Secondary Inlet (from CT basin)
 - o T-8: Heat Exchanger – Hot Secondary Outlet (to CT header)
 - o $\Delta 5/6$: Heat Exchanger
 - o $\Delta 2/4$: Core – $13.8^\circ\text{F} = 1$ MW at 500 GPM steady state (*calibration point for N-16*)
- RADIATION MONITORING AND EVAC PANEL (*reference schematic*):
 - o Area Monitors (*trip setpoints for EVAC*) - Over the Pool (100 mr/hr); Control Room (5 mr/hr); Demin; West Wall (100 mr/hr)
 - o Effluent Monitors (*trip setpoints for EVAC*) – Stack pump control; Stack Gas (2500 CPM); Stack Particulate (70000 CPM); Filter GM (15000 CPM); AUX GM (15000 CPM)
 - o Radiation Chart Recorder – rad traces; CAM (fuel failure); Pool Resistivity (>0.5 MOhm-cm); Pool PH (5.5-7.5)

REACTOR SYSTEMS WALKDOWN LAB – PART BREACTOR PLANT SYSTEMS:3RD FLOOR HVAC ROOM (REFERENCE SCHEMATIC):

- Purpose and functioning of a confinement system
- Inlet plenum from outside (~180 CFM)
- Control Room AC Recirc (nominal 520 CFM)
- Bay AC Recirc (nominal 5575 CFM)
- Bay Main Exhaust Fan (nominal 1400 CFM) & Filters – w/ Filter GM
- Confinement Fans #1 & #2 (nominal 600 CFM ea.); disconnects; dampers
- Iso-kinetic Probes (normal and confinement flows)
- Stack Gas Pump w/ Filter Particulate Monitor (scintillator) and Stack Gas Detector (GM); AUX detector (GM)

POOLTOP

- 3 functions of primary water: coolant, moderator, radiation shield
- 4 control rod drives and magnets (*only 3 scrammable*); up/down limit switches; SDM limit switches (below 11.3” >4W); REG drive >13.5” limit switch for AUTO mode.
- Location of T-2 (SCRAM @ 116°F); Pool temp switch (trip at 114°F +/- 2°F)
- PH and Conductivity monitors; pool level monitor (bubbler) and measuring stick; evaporation
- Core layout; location of NI’s; beamports & TC; fuel storage

MER (REF PRIMARY SCHEMATIC)

- Primary (500 GPM) and Secondary (700 GPM) Pumps w/ VFD controllers
- RTD’s #5-#8 + transmitters in cabinet
- RTD #9 – Cooling system control point (90°F primary return temp controls CT fan VFD)
- Plate and Frame SS Heat Exchanger: 126 plates – counterflow arrangement
- Secondary Makeup water
- Primary and Secondary Isolation Valves
- Cooling Tower Fan VFD
- Motor Control / Load Distribution Panel SB-1

PPV (REF DELAY TANK DRAWINGS - DO NOT GO DOWN LADDER INTO PIT)

- Location and function of delay tanks (~1400 gallons total) w/ shielding wall
- RTD’s #3 & #4
- Demineralizer skid w/ connections to primary
- Annubar Flow Meter w/ monometer (flow scram)
- Waste water handling

COOLING TOWER YARD (REF SECONDARY SCHEMATIC)

- Secondary water chemistry control system – corrosion and scale inhibitors; biocide
- Secondary poly-disk filter for particulates – demonstrate cleaning mode
- PURPOSE of secondary filtration/chemistry control – prevent fouling/corrosion of heat exchanger/ CT baffles
- Tower – open side access door to view basin sweeper; fan; inlet header; baffles

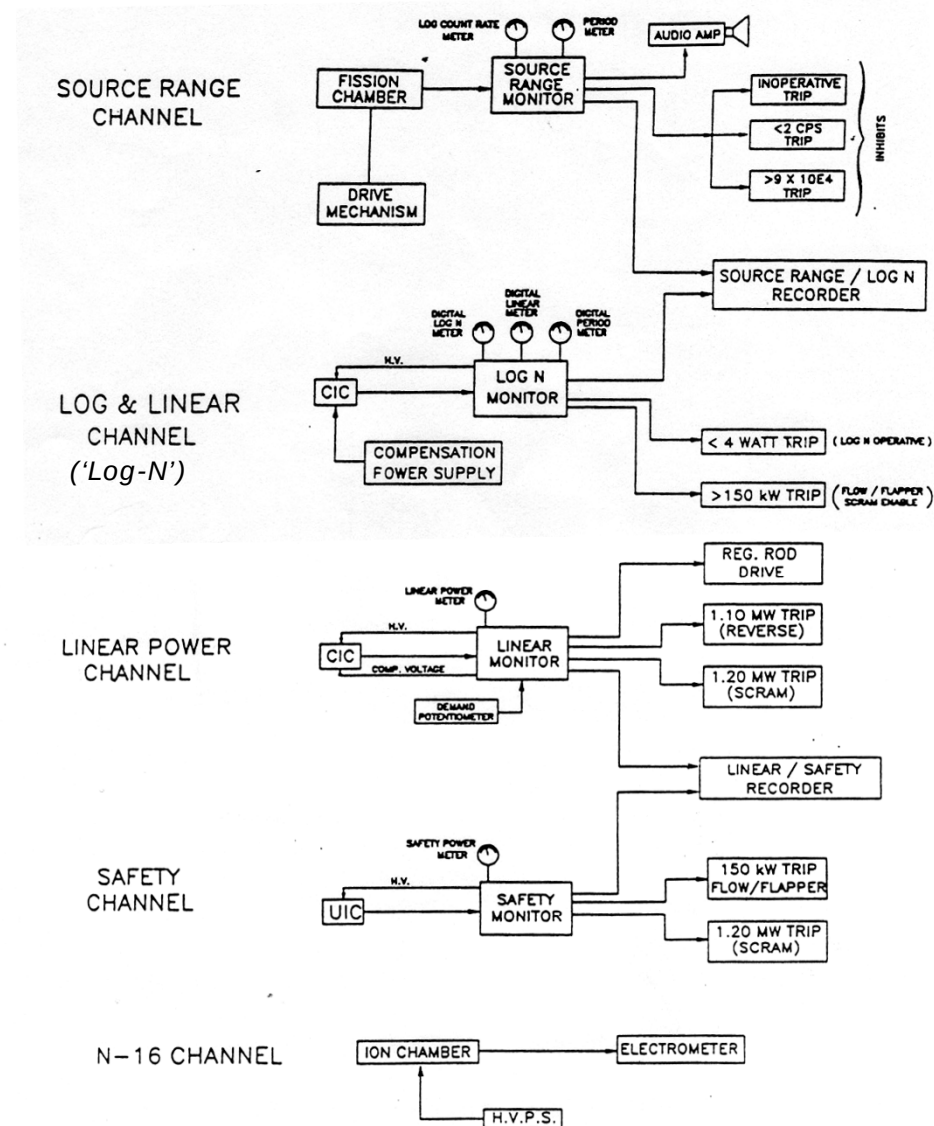
Reactor Instrumentation & SCRAM System

Seven Reactor SCRAM's:

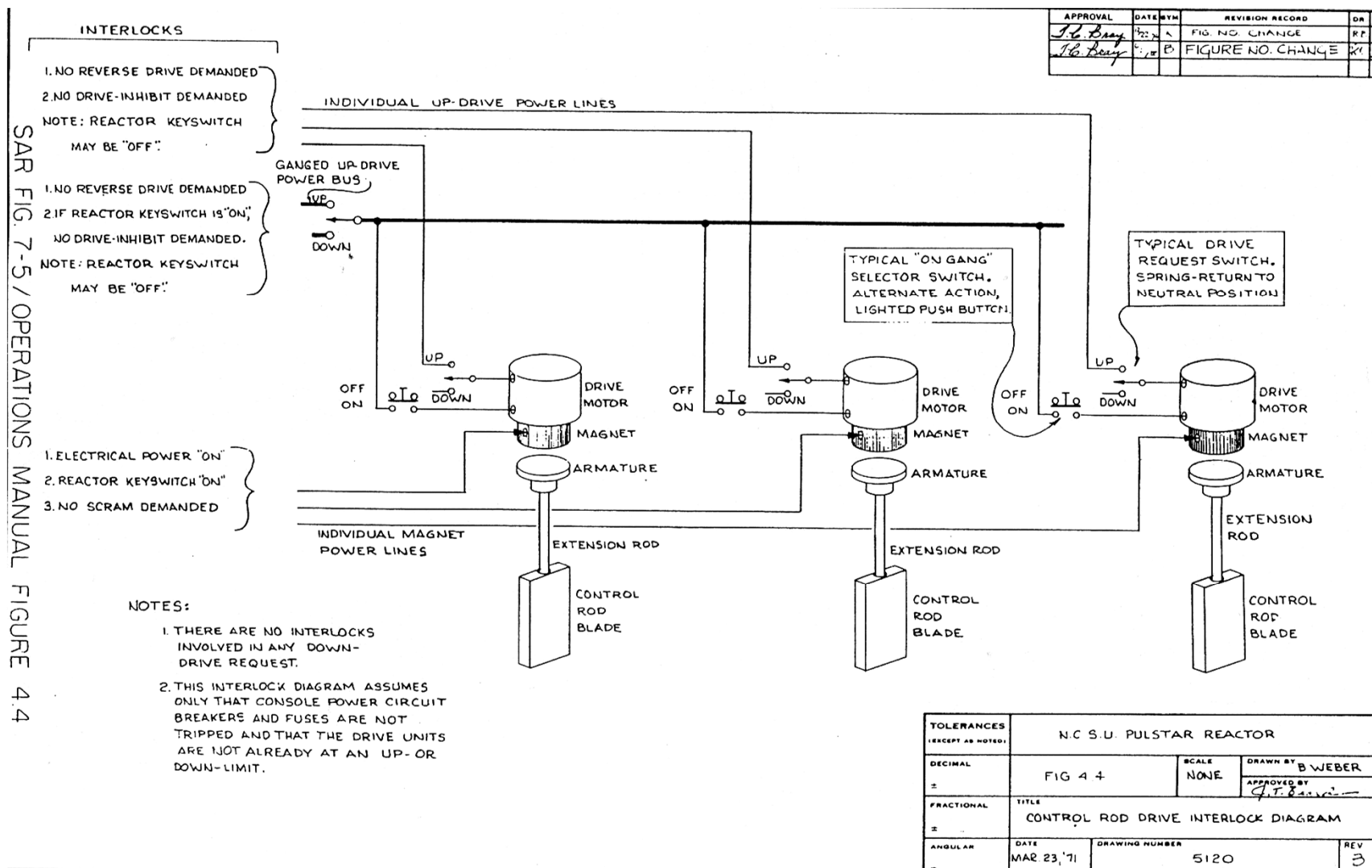
1. Linear Power Channel SCRAM at $\geq 120\%$ Full Power (1MW)
2. Safety Power Channel SCRAM at $\geq 120\%$ Full Power (1MW)
3. Primary Coolant Flow SCRAM at ≤ 475 GPM (*enables if Log-N or Safety Channel ≤ 150 kW*)
4. Flow Flapper Valve OPEN SCRAM (*enables if Log-N or Safety Channel ≤ 150 kW*)
5. Pool Level SCRAM at $\leq 36"$ below full
6. Pool Temperature SCRAM if T_2 (rtd) $\geq 116^\circ\text{F}$
7. Manual SCRAM – Depress Red Button on main console

Plus:

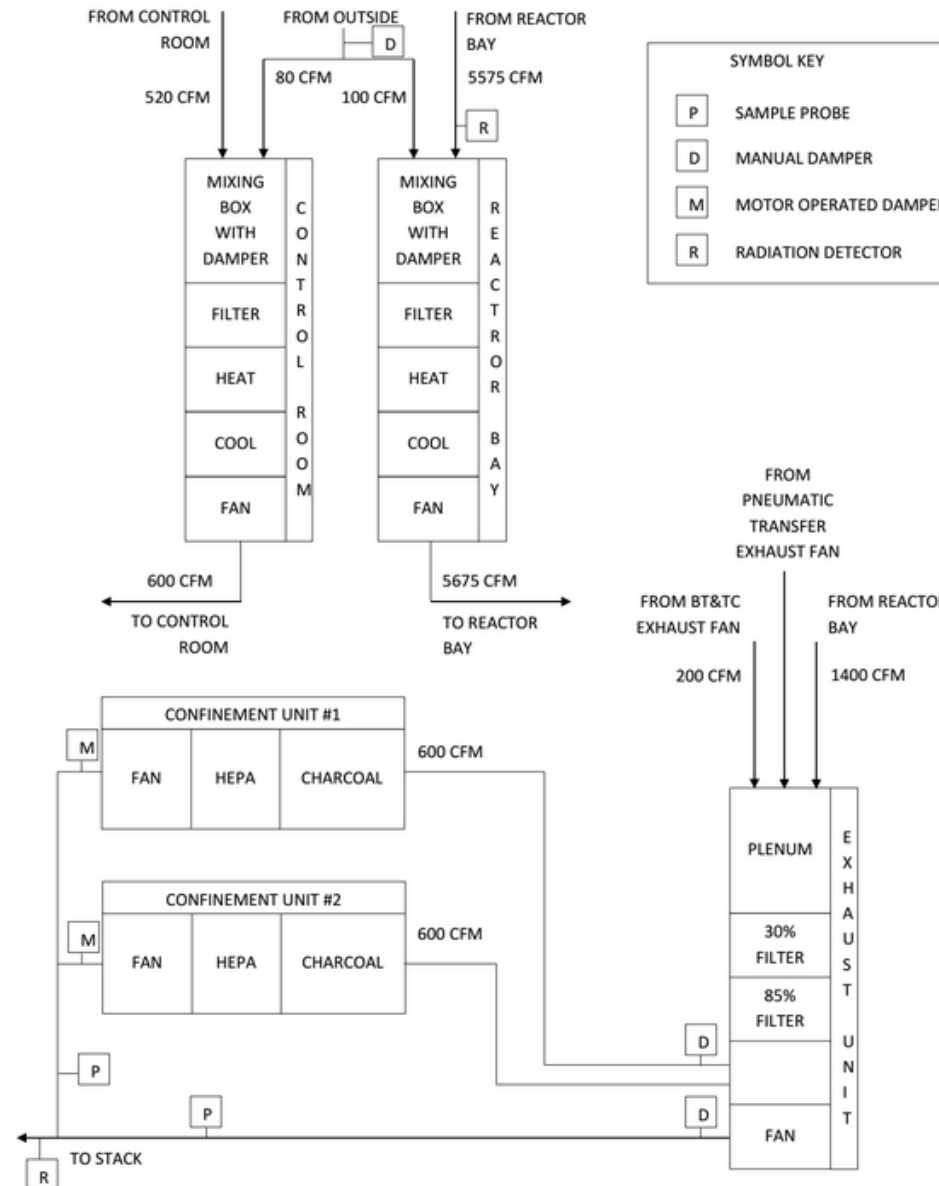
- Console Keyswitch
- SCRAM Logic Ground Fault



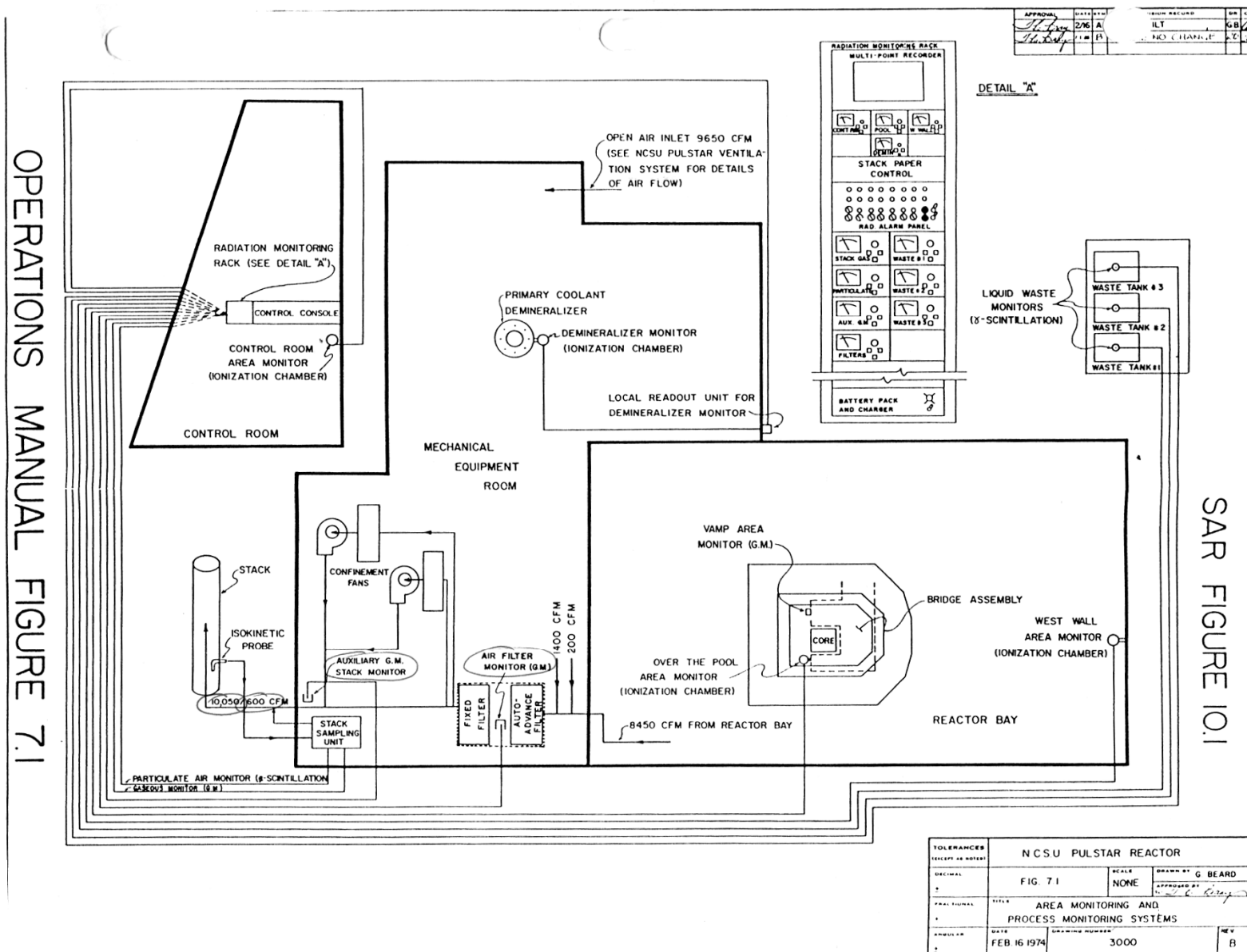
Control Rod – Electromagnet Assembly



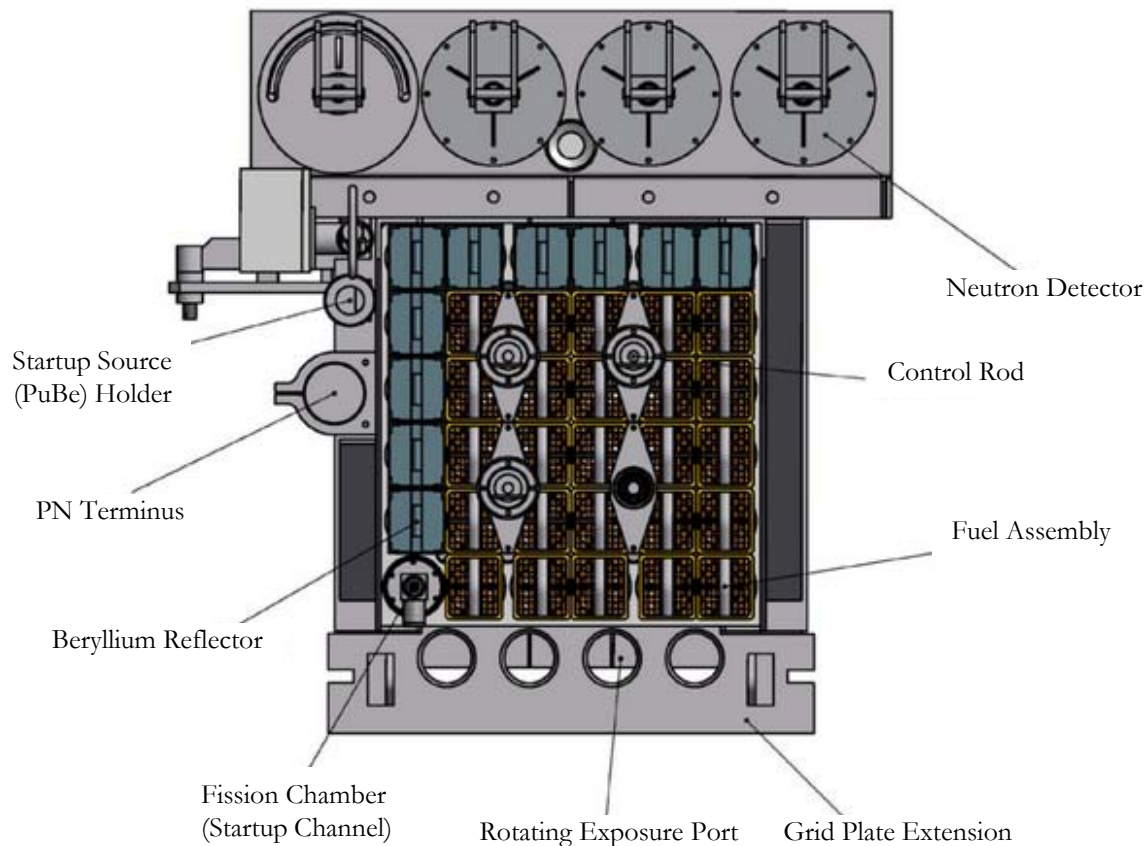
PULSTAR Confinement & HVAC Systems



PULSTAR Radiation Monitoring and EVAC System



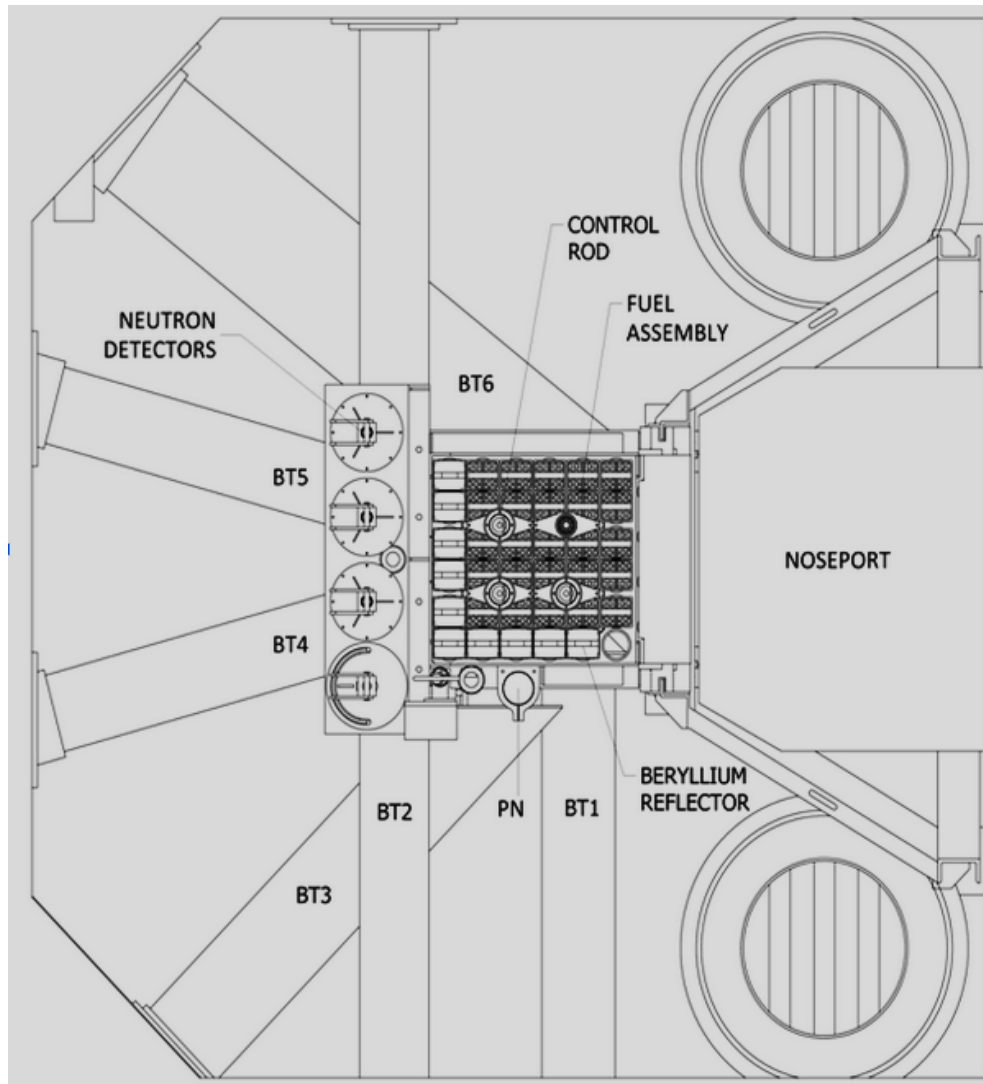
PULSTAR – Core and Gridplate Assembly



Core Structures:

- 36 Position Grid Plate
- 25 Fuel Assemblies
- 10 Reflector Elements
- 4 Control Rods
- 3 NI Power Monitors
- Cooling Plenum
- Flapper Valve Assembly
- Startup Source
- Fission Chamber

In-Pool Irradiation and Beam Port Facilities



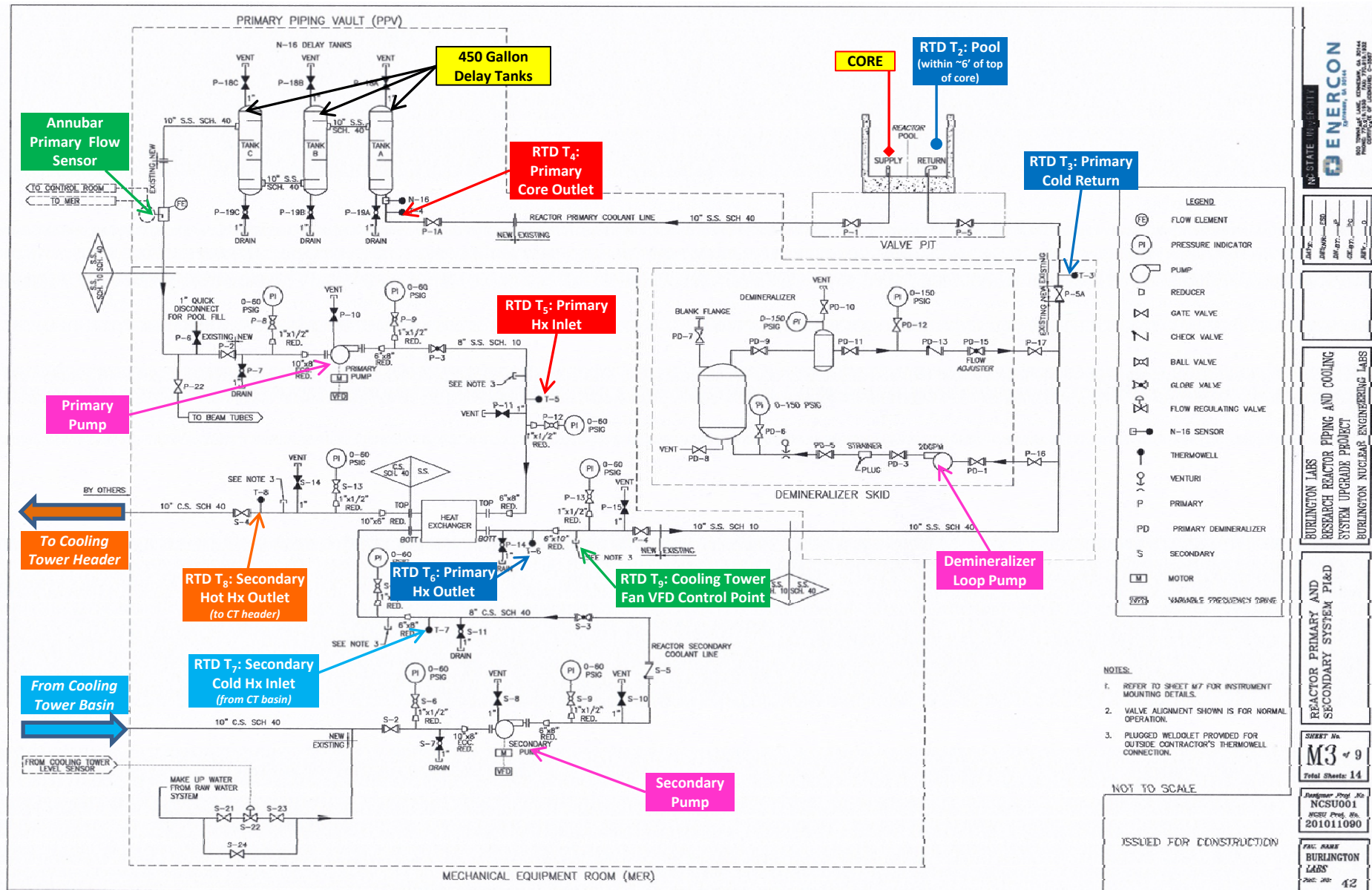
Beam-Port Facilities:

- BP#4 - Neutron Diffractometer
- BP#5 - Neutron Imaging Facility
- BP#6 - Positron Intense Beam Fac.
- Noseport - UCN Source

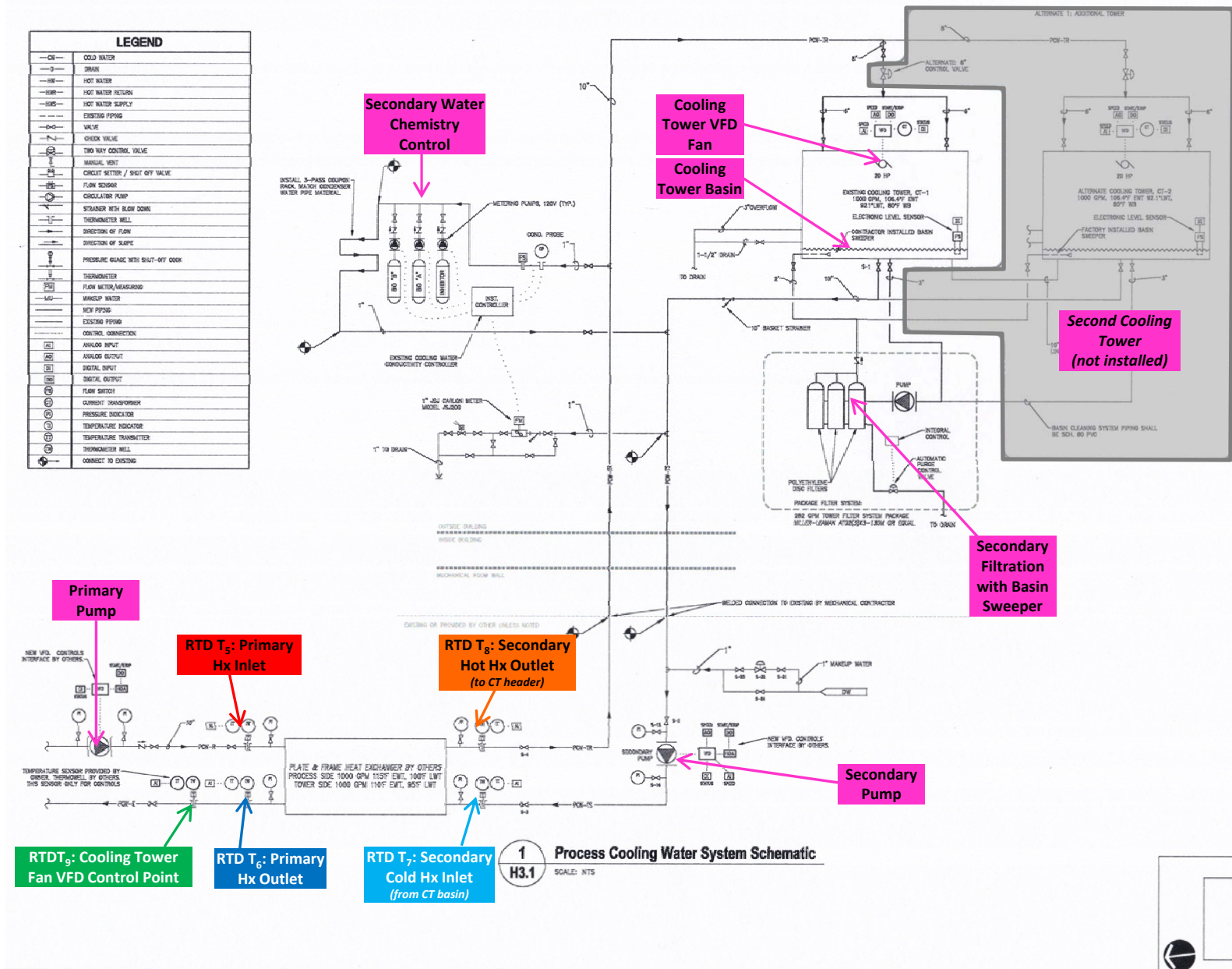
In-Pool Irradiation Facilities:

- 4 Rotating Exposure Ports
- 2 Dry Tubes
- 1 PN Terminus (ex-core)
- 3.5" I.D. Standpipes

PULSTAR - Primary Cooling System Schematic



PULSTAR - Secondary Cooling System Schematic



NC STATE UNIVERSITY
EDMONSON ENGINEERS
1000 W. MAIN ST., SUITE 100
Raleigh, NC 27601

DATE: 12/20/10
DESIGNER: TAC
DRAWN BY: TAC
CHECKED BY: JUE

NC STATE UNIVERSITY
BURLINGTON LABS
NUCLEAR REACTOR COOLING

COOLING P&ID

SHEET No.
H3.1
Title Sheet

Designer Proj. No.
12-027
NCSU Proj. No.
201011090

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